# HW#1 Real-time Analysis of a HW-SW Platform

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# Homework Goal

- In this homework, you will add profiling hardware to the Aquila core to analyze the program execution behavior
  - Learn how to profile a program on the real platform
  - Use the Xilinx Integrated Logic Analyzer (ILA) for debugging
  - The CoreMark benchmark program is used here
- Deadline: 10/07, 17:00
  - Upload your source code and a two-page report to E3
    - Only PDF file is allowed for report submission.
  - The TAs will set up a schedule for you to demo your code

# Synthesis-Execution Flow

- To run the HW-SW system on an FPGA, you must follow the steps:
  1. Generate the HW bit file
  2. Power on the Arty board
  3. Use a terminal program to connect to the FPGA via UART
  4. Program the FPGA
  5. Send an ELF program to the FPGA via UART
  6. Wait for the program to execute and print results

# Circuit Implementation for FPGAs

# Synthesize Aquila (No Cache, FPU)

aquila_ap - [D:/xilinx/aquila/aquila_build/aquila_ap/aquila_ap.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Quick Access write_bitstream Complete

Default Layout

Flow Navigator

- SIMULATION
  - Run Simulation
- RTL ANALYSIS
  - Run Linter
  - Open Elaborated Design
- SYNTHESIS
  - Run Synthesis
  - Open Synthesized Design
- IMPLEMENTATION
  - Run Implementation
  - Open Implemented Design
- PROGRAM AND DEBUG
  - Generate Bitstream
  - Open Hardware Manager
    - Open Target
      - Program Device
      - Add Configuration Mem

PROJECT MANAGER - aquila_ap

Sources

- Clock_Generator : clk_wiz_0 (clk_
- Aquila_SoC : aquila_top (aquila_top
  - RISCV_CORE0 : core_top (core
  - TCM : sram_dp (sram_dp.v)
  - CLINT : clint (clint.v)
  - ATOM_U : atomic_unit (atomic_
- UART : uart (uart.v)

Hierarchy IP Sources Libraries

Source File Properties

soc_top.v

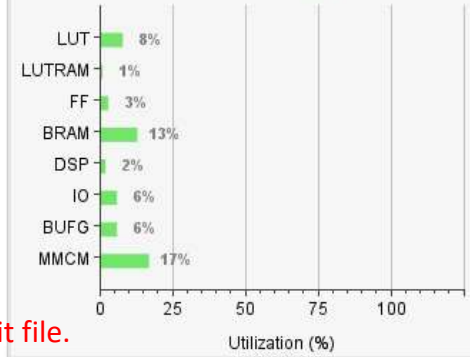
General Properties

Project Summary

Overview | Dashboard

zation Post-Synthesis | Post-Implementation

Graph | Table




Power

| | |
|---|---|
| Total On-Chip Power: | 0.246 W |
| Junction Temperature: | 26.1 °C |
| Thermal Margin: | 58.9 °C (12.8 W) |
| Effective ϑJA: | 4.6 °C/W |
| Power supplied to off-chip devices: | 0 W |
| Confidence level: | Medium |

Implemented Power Report

Tcl Console Messages Log Reports Design Runs

| Name | Constraints | Status | WNS | TNS | WHS | THS | WBSS | Methodology | LUT | FF | BRAM | URAM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| synth_1 (active) | constrs_1 | synth_design Complete! | | | | | | | 4237 | 324 | 16 | 0 |
| impl_1 | constrs_1 | write_bitstream Complete! | 5.364 | 0.000 | 0.050 | 0.000 | | 36 Warn | 4784 | 411 | 17 | 0 |

Click this to generate the FPGA bit file.

Click this to connect to the FPGA board.

# Program The FPGA

aquila_ap - [D:/xilinx/aquila/aquila_build/aquila_ap/aquila_ap.xpr] - Vivado 2025.1

File Edit Flow Tools Reports Window Layout View Help Quick Access write_bitstream Complete

Dashboard Default Layout

Flow Navigator

SIMULATION
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Open Implemented Desig

PROGRAM AND DEBUG
Generate Bitstream
Open Hardware Manager
1 Open Target
Program Device
Add Configuration Men

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210319B7CA36A

There are no debug cores. Program device Refresh device

Hardware 2

| Name | Status |
|---|---|
| localhost (1) | Connected |
| xilinx_tcf/Digilent/210319B7CA36 | Open |
| xc7a100t_0 (1) | Not programm |
| XADC (System Monitor) | |

Properties

Select an object to see properties

Tcl Console Messages Serial I/O Links Serial I/O Scans

```
INFO: [Labtoolstcl 44-466] Opening hw_target localhost:3121/xilinx_tcf/Digilent/210319B7CA36A
set_property PROGRAM.FILE {D:/xilinx/aquila/aquila_build/aquila_ap/aquila_ap.runs/impl_1/soc_top.bit} [get_hw_devices xc7a100t_0]
current_hw_device [get_hw_devices xc7a100t_0]
refresh_hw_device -update_hw_probes false [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1435] Device xc7a100t (JTAG device index = 0) is not programmed (DONE status = 0).
```

Type a Tcl command here

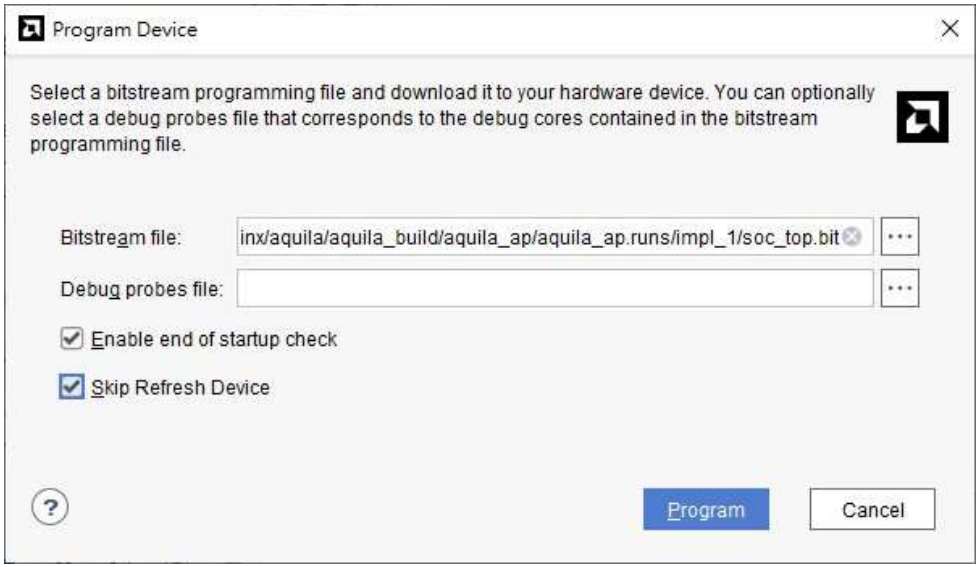
After connecting to the Arty board, click this to program the circuit into the FPGA.

But before you program it, you should have a terminal window connects to the correct UART port first.

Power on the Arty board, and click this to connect to the board. Select “auto connect” In the pop-up dialog box.

# Select the Bit File for Programming

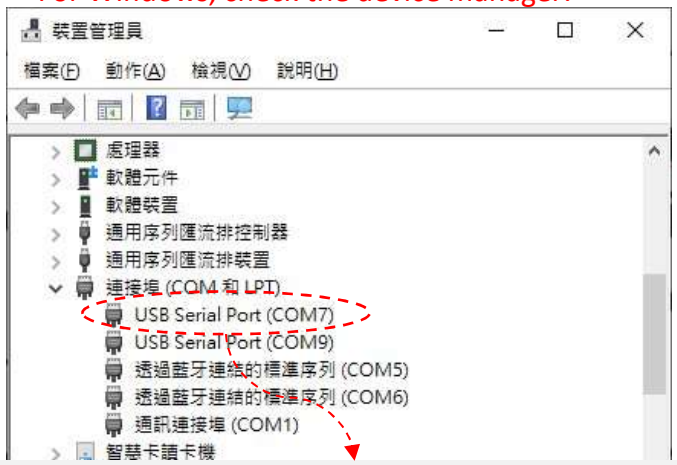
- The FPGA bit file, `soc_top.bit`, is located under `aquila_ap/aquila_ap.runs/impl_1/` after circuit implementation
- After clicking the “Program device” hyperlink, a file browser will pop-up to let you to load the bit file:



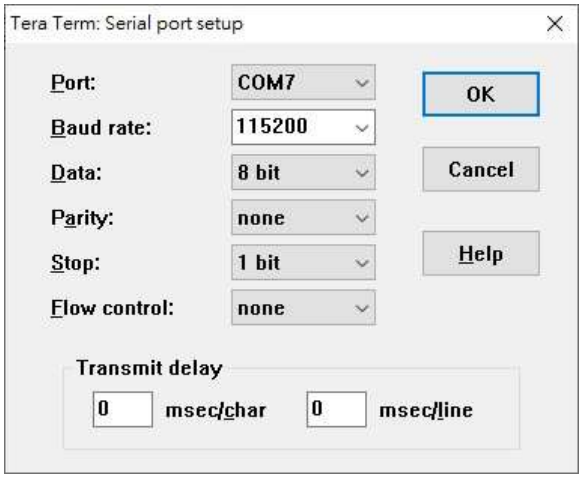
# Terminal Settings for Windows

- You can use the TeraTerm† on Windows, or GTKTerm on Linux as an I/O terminal of the Aquila SoC in FPGA
  - Please do not use “minicom” on Linux!
- To connect the terminal program to Aquila, you must set the right COM port and UART parameters:

For Windows, check the device manager:




Use the COM port that shows up when you plug-in the Arty board. There may be two new ports show up. One is for JTAG, one is for UART. You have to try to see which is which!




† https://ttssh2.osdn.jp/index.html

# Terminal Settings for Linux

- For Linux, use "`sudo dmesg`" to see the Arty devices:

```
[2064339.294090] usb 1-7: Product: Digilent USB Device
[2064339.294091] usb 1-7: Manufacturer: Digilent
[2064339.294092] usb 1-7: SerialNumber: 210319A8C7F1
[2064339.297789] ftdi_sio 1-7:1.0: FTDI USB Serial Device converter detected
[2064339.297801] usb 1-7: Detected FT2232H
[2064339.297928] usb 1-7: FTDI USB Serial Device converter now attached to ttyUSB0
[2064339.299920] ftdi_sio 1-7:1.1: FTDI USB Serial Device converter detected
[2064339.299928] usb 1-7: Detected FT2232H
[2064339.300026] usb 1-7: FTDI USB Serial Device converter now attached to ttyUSB1
```

Arty JTAG programming device.

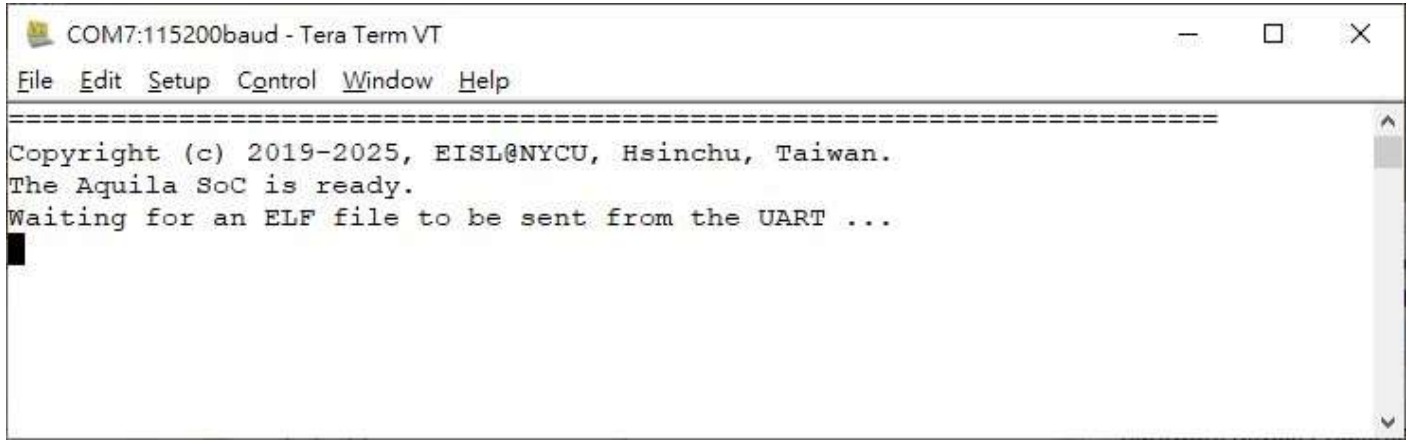
Arty serial port device.

- GTKTerm configuration:

# Run Applications on the Arty Board

# Running an Executable on the FPGA

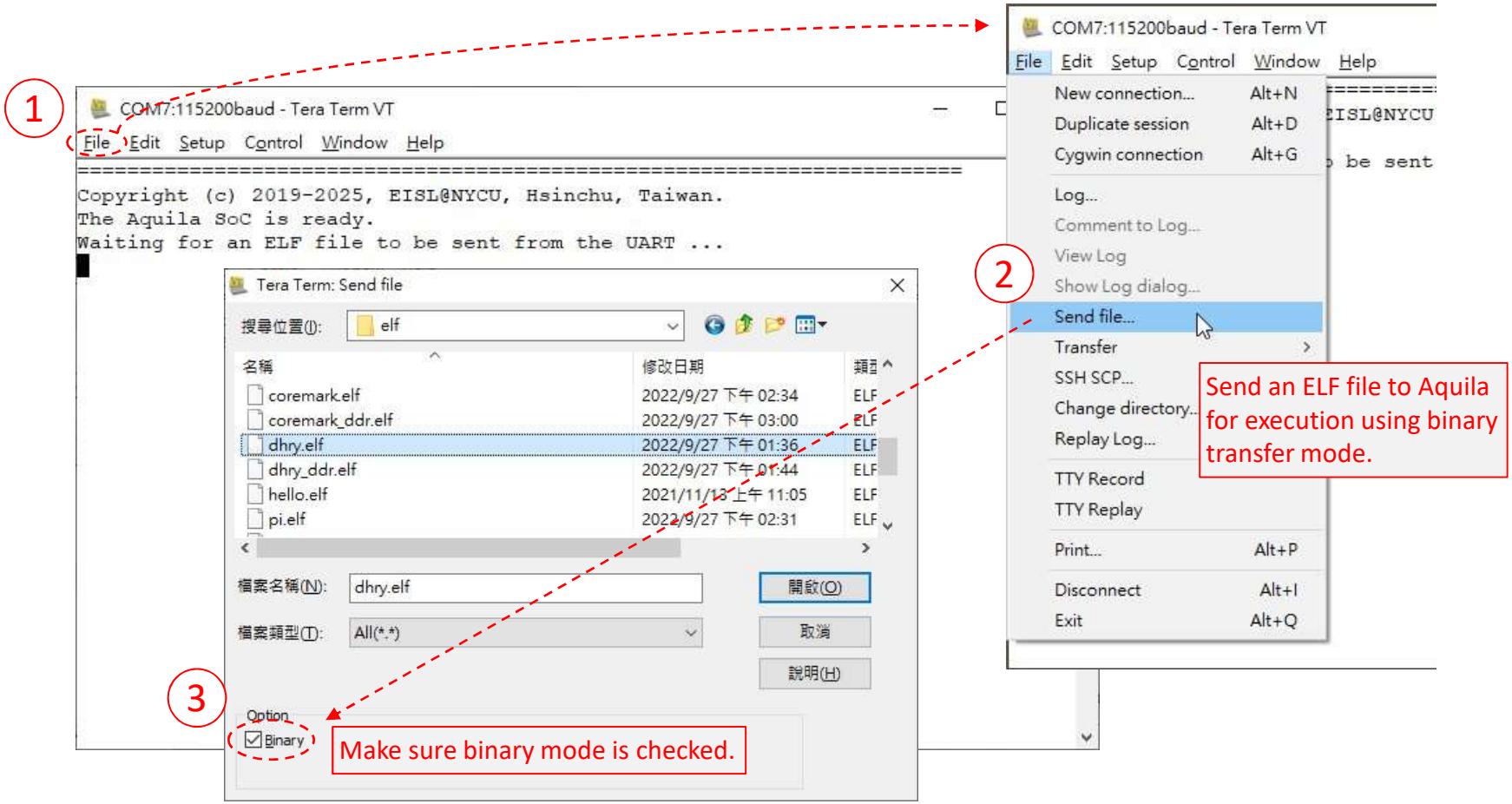
- Once the FPGA is programmed, a message is shown on the terminal program, waiting for an ELF file to be send to the Aquila SoC for execution:




- Under Linux, you can send an ELF file to FPGA with the shell command: `$cat dhry.elf > /dev/ttyUSB1`

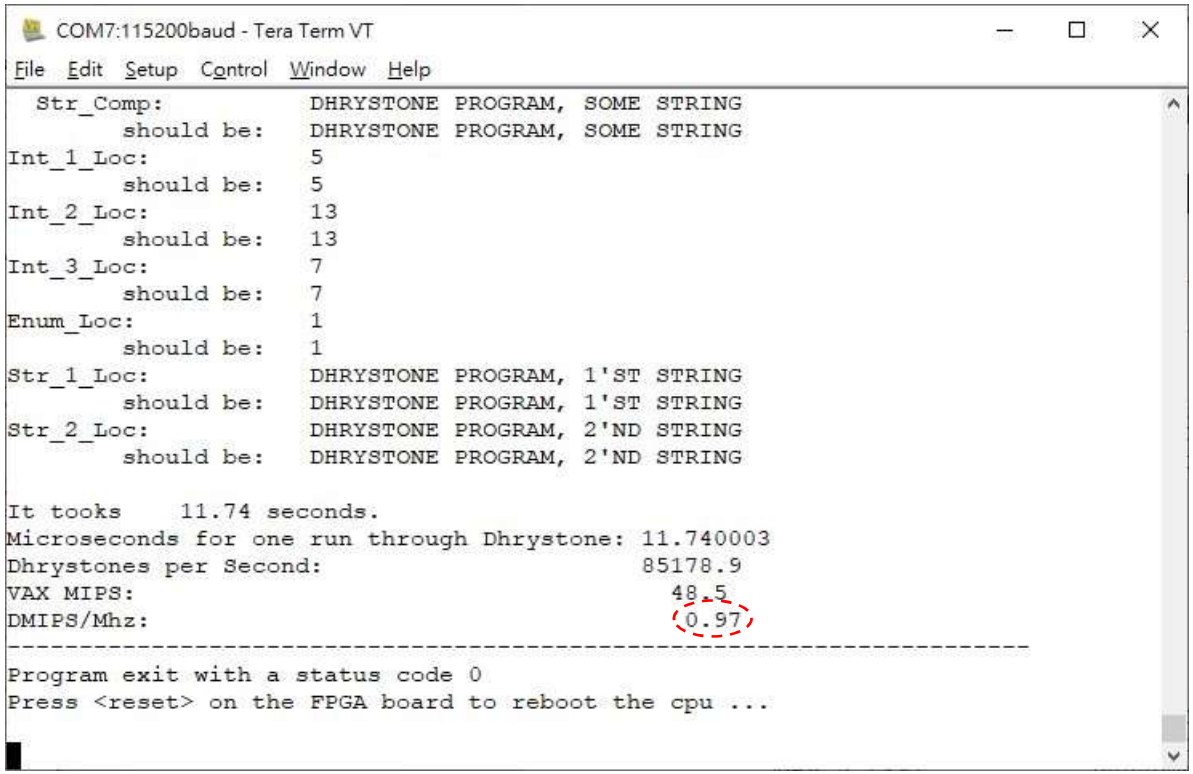
# Sending a *.elf to FPGA (Windows)

- For Windows (TeraTerm), use the “Send file” menu:



# Dhrystone Result

- Send `dhry.elf` to Aquila and you should see:
  - Note: the actual DMIPS number depends on how well you optimize the `elibc/string.c` functions in HW#0



# Analyze Aquila Execution on FPGA

- To analyze the behavior of the Aquila SoC running a program on the real FPGA board, you can use the Integrated Logic Analyzer (ILA)

- Real-time ILA circuit probing steps:
  - Embed signal probes into your circuit (the Aquila SoC here)
  - Set a trigger condition to capture signal traces to on-chip RAM
  - Perform a post-mortem analysis on a PC afterwards

- A tutorial on ILA is in the Appendix (page 31– ).

# Profiling the Target Benchmark

# CoreMark

- For this HW, we will use CoreMark† as the target:

COM7:115200baud - Tera Term VT

File Edit Setup Control Window Help

```
2K performance run parameters for coremark.
CoreMark Size    : 666
Total ticks      : 47339
Total time (secs): 0.047339
Iterations/Sec   : 105.621158
Iterations       : 5
Compiler version : GCC15.1.0
Compiler flags   : -Wall -O3 -I../elibc -fno-builtin -march=rv32ima_zicsr_zifenc
ei -mabi=ilp32 -mstrict-align -fno-inline-small-functions -DITERATIONS=5 -DUSE_C
LOCK=1
Memory location  : HEAP
seedcrc          : 0x0000e9f5
[0]crclist       : 0x0000e714
[0]crcmatrix     : 0x00001fd7
[0]crcstate      : 0x00008e3a
[0]crcfinal      : 0x0000f24c
Correct operation validated. See README.md for run and reporting rules.
CoreMark 1.0 : 105.621158 / GCC15.1.0 -Wall -O3 -I../elibc -fno-builtin -march=r
v32ima_zicsr_zifencei -mabi=ilp32 -mstrict-align -fno-inline-small-functions -DI
TERATIONS=5 -DUSE_CLOCK=1 / HEAP
-----------------------------------------------------------------------
Program exit with a status code 0
Press <reset> on the FPGA board to reboot the cpu ...
```

† https://github.com/eembc/coremark

# Profiling a Program for Optimization

- Often, 80% of the program execution time resides in 20% of the code → how do you find the hotspots?

- An easy way to do this is to use a software profiler:
  - Sampling-based hot spot analysis
    - Insert a timer interrupt service routine (ISR) into your program
    - Interrupt the processor at a fixed frequency, say, 100 Hz, and the ISR collects samples of the PC during execution
    - The PC samples are used to estimate the time spent in a function
  - Counter-based call analysis
    - Insert counter code into every function
    - Record the caller and calling frequency

# Example: Profiling on PC

- In Linux, GCC can be used to profile a program:

```
$ gcc -O2 -pg  my_prog.c -o my_prof
$ ./my_prog
$ gprof ./my_prog gmon.out > profile.txt
```

  - A program compiled with the `-pg` flag will have profiling code inserted into the program
  - When executed, the profiling code will collect the runtime information and store it to the binary file `gmon.out`
  - The command `gprof` can convert `gmon.out` to a text file
  - Note that if you use WSL, only WSL 2.0 works. For WSL 1.0, `gprof` only provide call analysis, not hotspot analysis
    - Check your WSL version using dos command: `WSL -l -v`

# Building Coremark with Makefile.linux

- We have created a Makefile.linux for you to build a gprof-enabled version of Coremark for Linux
- Under `aquila_sw/CoreMark`, simply type the following commands:

```
$ make -f Makefile.linux
$ ./coremark
$ gprof ./coremark gmon.out > profile.txt
```

the runtime statistics of hotspots will be in profile.txt

# The Profile of CoreMark

- With Ubuntu22.04/WLS2, gcc 11.4.0, the statistics looks like:

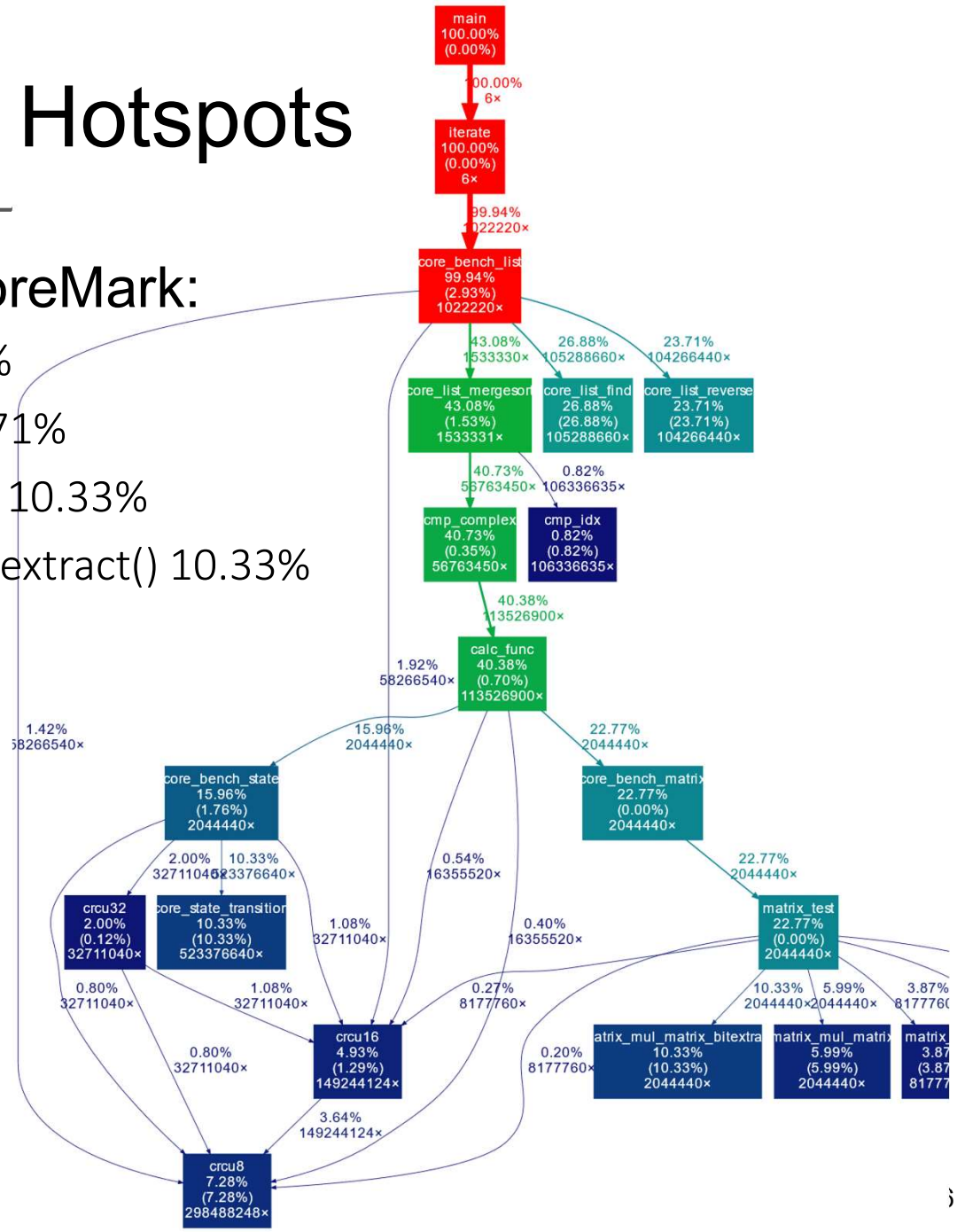
```
Flat profile:
Each sample counts as 0.01 seconds.
%   cumulative   self              self     total
time   seconds   seconds    calls   s/call   s/call  name
26.88      2.29     2.29 105288660     0.00     0.00  core_list_find
 23.71      4.31     2.02 104266440     0.00     0.00  core_list_reverse
 10.33      5.19     0.88 523376640     0.00     0.00  core_state_transition

                                  . . . . .

                           Call graph (explanation follows)
granularity: each sample hit covers 4 byte(s) for 0.12% of 8.52 seconds
index % time    self  children    called     name
                                                 <spontaneous>
[1]    100.0    0.00    8.52                 main [1]
                0.00    8.52       6/6           iterate [2]
                0.00    0.00       1/1           core_list_init [23]

                                  . . . . .
```

- You can use `gprof2dot.py + graphviz`, to draw a call-graph from `profile.txt`:

```
$ gprof2dot.py profile.txt | dot -Tsvg -o coremark.svg
```

# Visualization of Hotspots

- Top 5 hotspots of CoreMark:
  - core_list_find()  26.88%
  - core_list_reverse() 23.71%
  - core_state_transition() 10.33%
  - matrix_mul_matrix_bitextract() 10.33%
  - crcu8() 7.28%



# Some Comments on using Gprof

- The result of gprof profiling depends on the gcc version and the processor you use
  - The true top-five hotspots for Aquila SoC may not be the same as the ones shown in the previous slide
- GCC tends to replace a small function by an inline function, which will mislead the profiling results
  - If a function is inlined, all calls to the function will be replaced by the direct insertion of the function into the caller
  - The GCC compiler flag `-fno-inline-small-functions` is used in the `Makefile` to avoid inlining

# Limitations of Software Profiler

- May not be available on the target platform
  - You need GCC+Glibc+Linux
  - We use GCC+elibc, which has no built-in profiling facility
- Timer interrupts can be obtrusive to the program, especially when the program has its own ISR
  - Storing profiling data in main memory can also be expensive
- Recursive functions cannot be profiled properly
  - Need better granularity
- Cannot differentiate between computations and memory/device accesses

# Profiling Using CPU Hardware

- Since we can modify the hardware of the CPU, can we design a CPU that automatically profile a program during execution?

- To count the execution cycles of a function, we can design a hardware counter that ticks whenever the PC is in that function
  - The starting address and the length of a function is in the *.map file from the compiler
  - We can further analyze the ratio of computation versus memory cycles that for a particular function
    - Note: you must take into account the stall cycles

# Warning on CoreMark Iterations

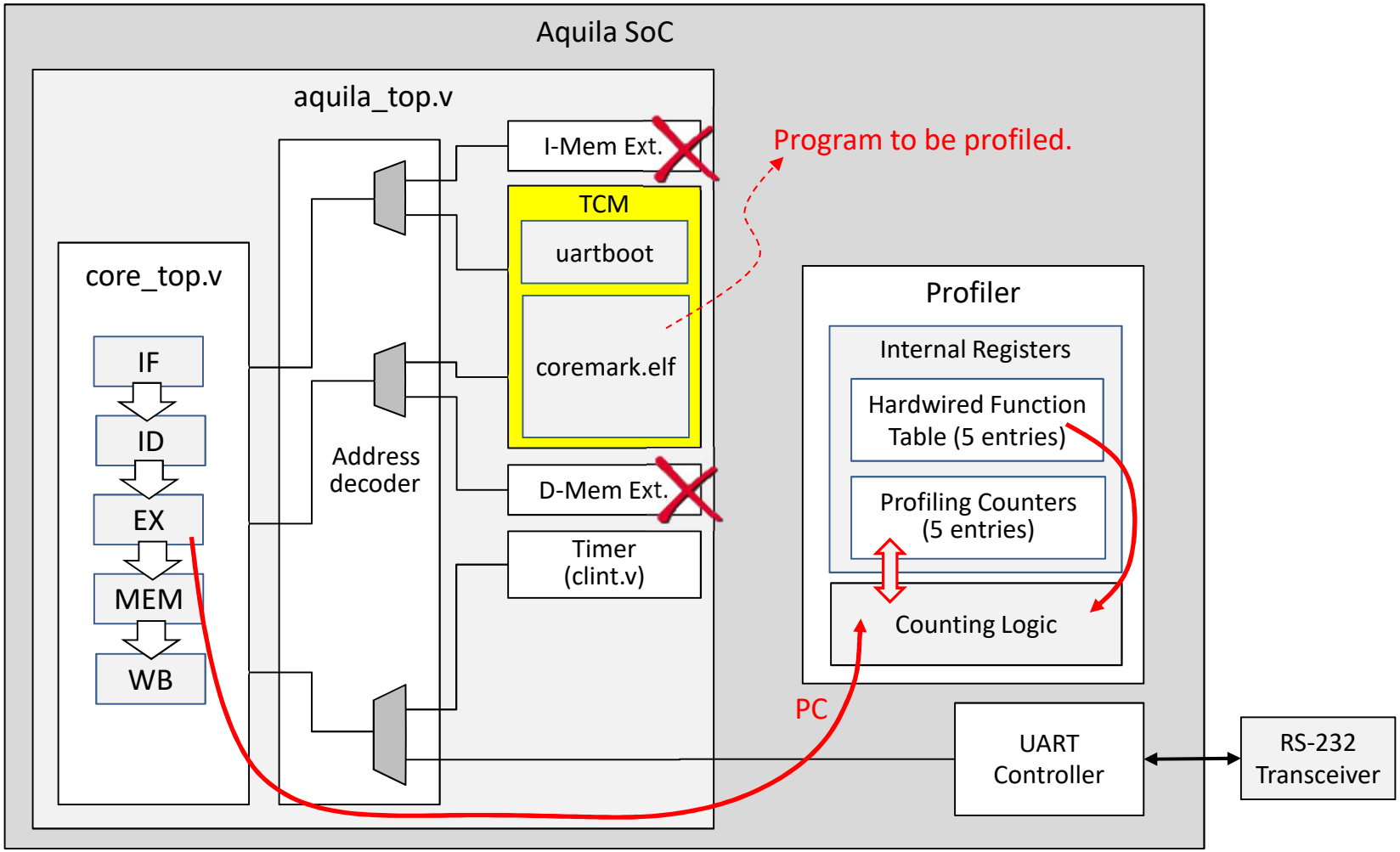
- There is a macro parameter in CoreMark that controls the # of iteration it runs
  - ITERATIONS=5 when the ELF is compiled for simulation
  - ITERATIONS=0 when it is for on-board execution
- When ITERATIONS=0, CoreMake will run as many as iterations such that the execution is longer than 10 sec
  - If the iteration number is too small, hotspot analysis will be bias towards printf() operations
- The parameter can be modified in Makefile

# Warning on Waveform Simulation

- Although this HW can be debugged with logic simulator, you shall run CoreMark on Arty to get the final statistics

- Two reasons why simulation gives you wrong results:
  - The UART controller has very different timing in simulation and in real-time execution
  - ITERATION=0 will take way too long to simulate

# Block Diagram of a CPU with Profiler



# How to Show the Counter Values

- There are two ways to show the counter registers values of the profiler:
  - Use ILA to capture the register values when the PC return from `main()` to `crt()` in the Coremark program.
  - Design a memory-mapped I/O (MMIO) interface to wrap around the counter registers of the profiler.

    You can refer to the interface design of `clint.v` for the example of a circuit block that has MMIO interface.
- For this homework, you can simply use the first method!
  - The 2nd method will be studied in HW#4 and used in HW#5.

# Your Homework

- Add HW code to Aquila such that it can collect the runtime profiling data for the top-five hotspots
  - You should also count the total CPU cycles and compute the CPU ratio of each function
  - You should calculate the ratio of computation versus memory cycles for each function
    - The cycles used to execute load/store instructions, including data memory stall cycles, are memory cycles
  - You can also try other more powerful profilers for analysis

- Write a 2-page double-column PDF report†:
  - Discuss what you have done to collect runtime statistics
  - Discuss what you have learned from the profiled data?

† A report template (of Microsoft Word format) is available on E3.

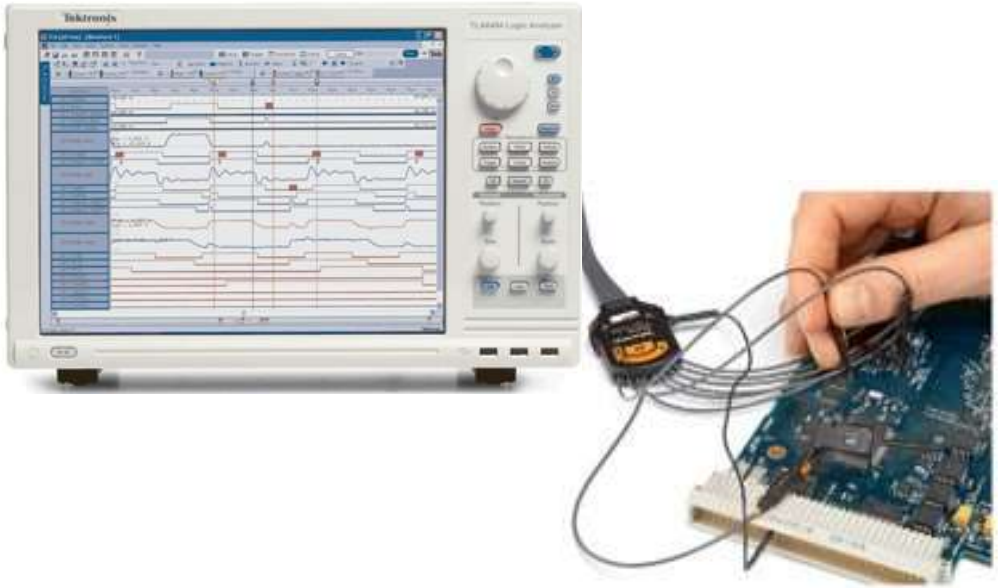
# Comments on Report Grading

- Your report will be graded using the following points:
  - Organization and writing style (25%)
  - The design of your profiling mechanism (25%)
  - Discussions on the profiling results (30%)
    - Comparison of the result to that on a PC
    - What you can say about the computation vs. memory cycles?
    - What you can say about the stall cycles?
  - Discussions on how to improve Aquila (20%)

# Appendix: ILA for Circuit Debugging

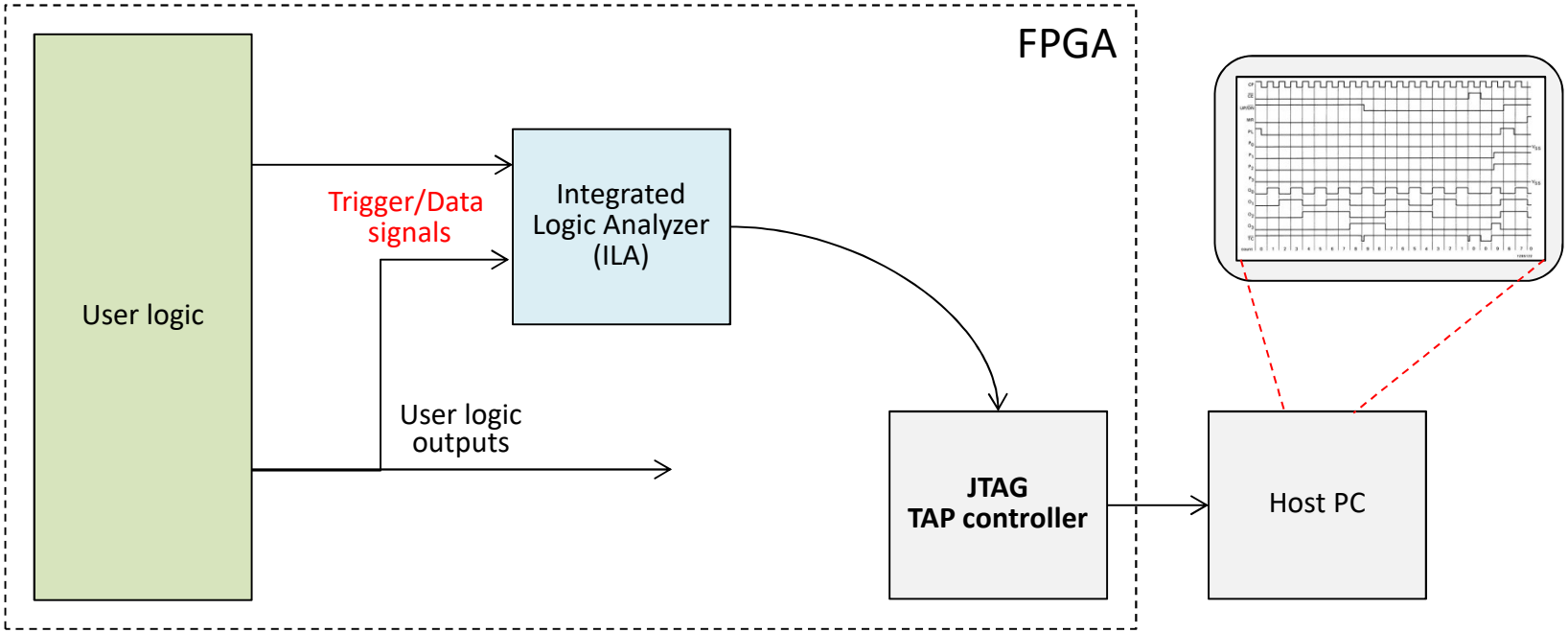
# Real-Time Probing Using Vivado

- Full-system simulations for complex logic and software behaviors would take too much time; and real devices are difficult to simulate
- In the good old days, for real-time debugging of a digital circuit, we use a logic analyzer for the job



# Vivado Integrated Logic Analyzer

- Vivado Integrated Logic Analyzer (ILA) is an IP that can be integrated into the hardware platform so that some signals in the user IP's can be intercepted and saved in a trace file for analysis



# Debug Your Circuit in Real-Time

- To debug your logic in real-time, you must “mark” the signals for debugging with one of the three methods:
  - Using the “synthesis attribute” syntax in Verilog-2001
  - Using the Vivado GUI IDE
  - Using the TCL command console (we don’t use TCL here)
- After marking the signals, you must set up the debug wizard so that ILA can capture the signals at runtime
- Do not mark the system clock. The waveform viewer has tick markers.

# Mark Debug Signals Using Verilog

- In Verilog-2001, you can set the synthesis attributes of a signal, for example:

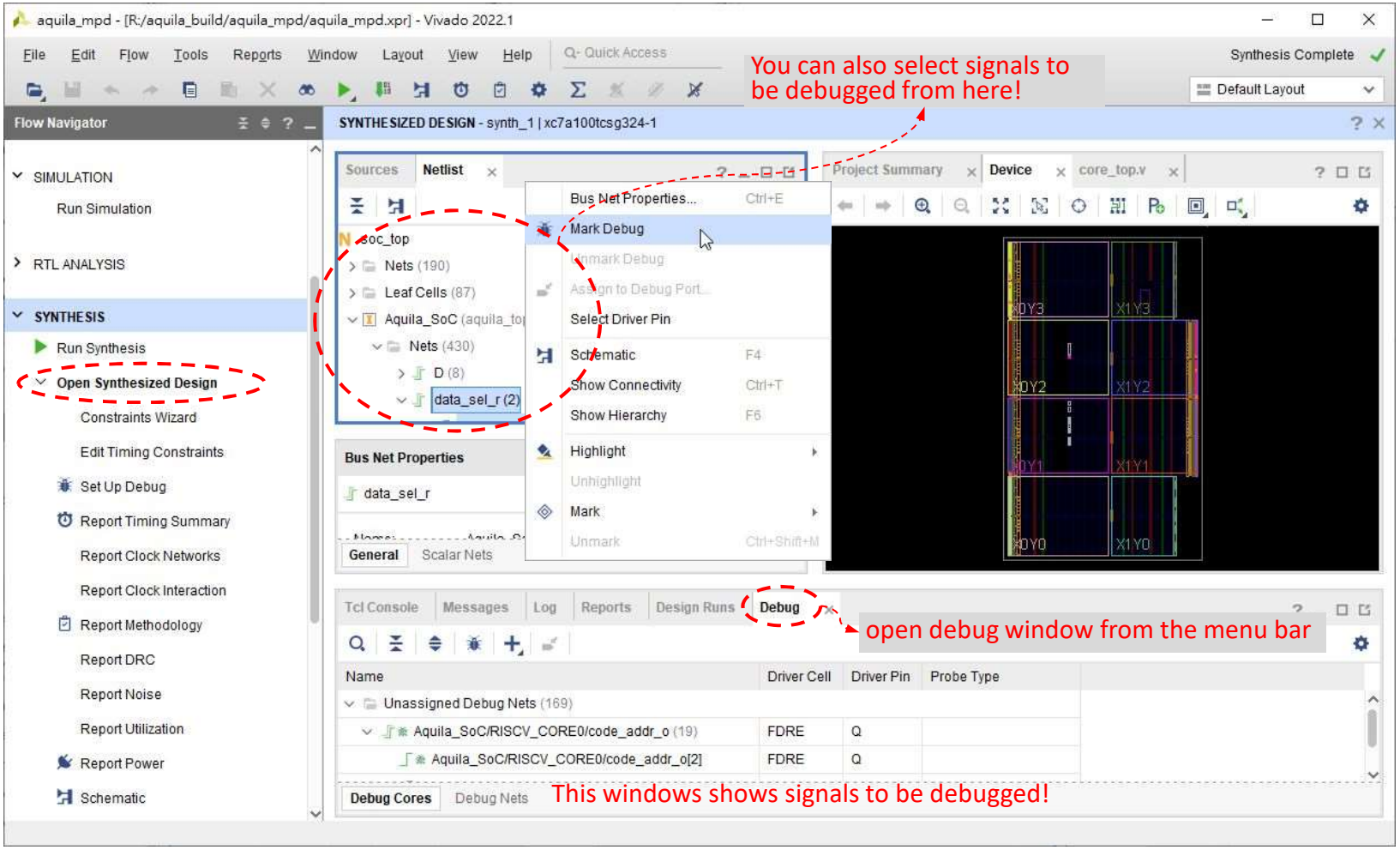
```
(* mark_debug = "true" *) wire my_signal
```

  This will turn on the “debug” attribute of `my_signal`.

- In Vivado, if your logic has signals with the debug attribute enabled, then:
  - The signals will not be “optimized-out” by the logic synthesizer, unless the signal is void
  - Vivado will insert an ILA IP into the synthesized design to monitor and capture these signals at runtime

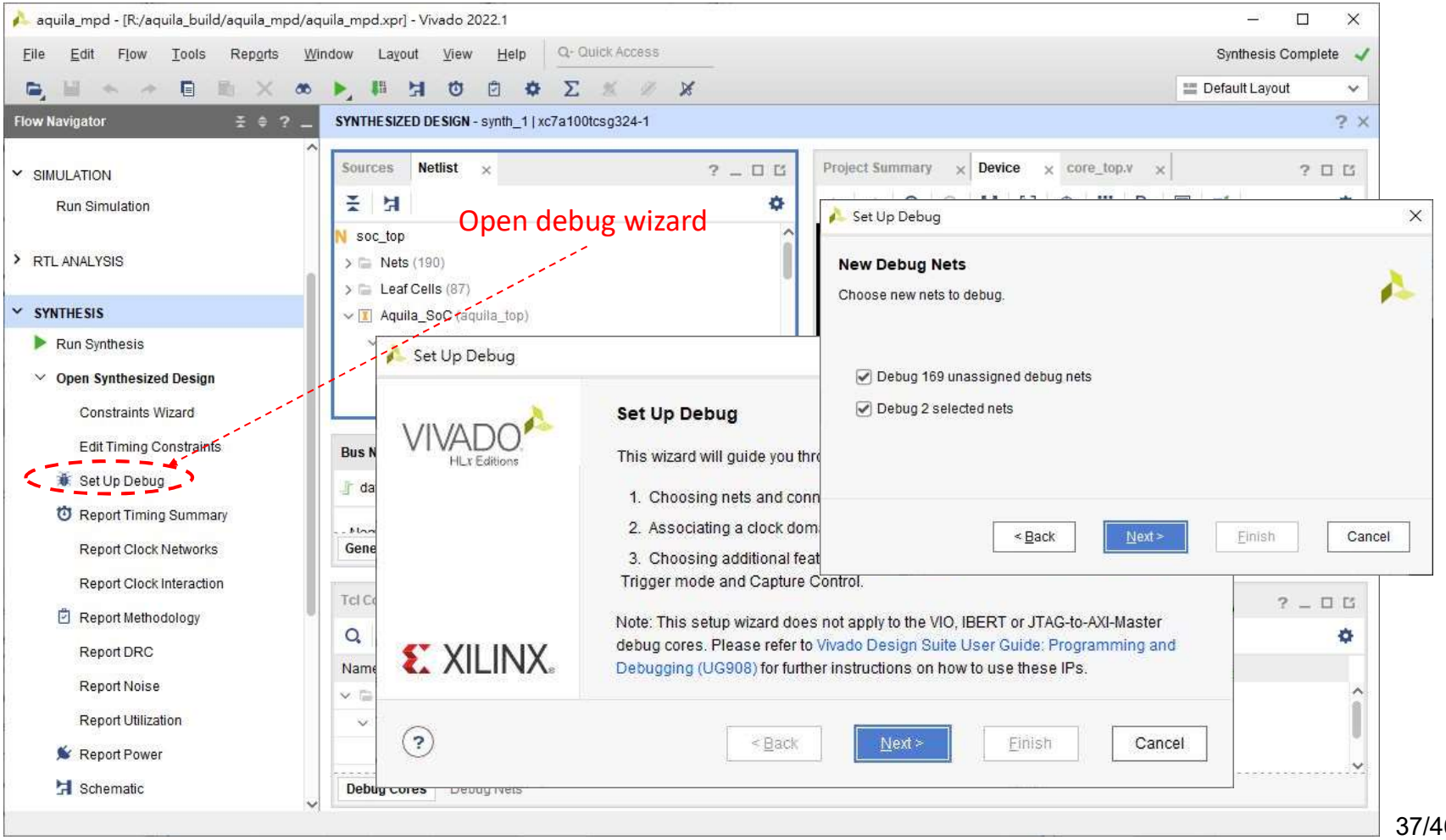
# Mark Debug Signals Using GUI

- To debug a circuit, open the synthesized design:



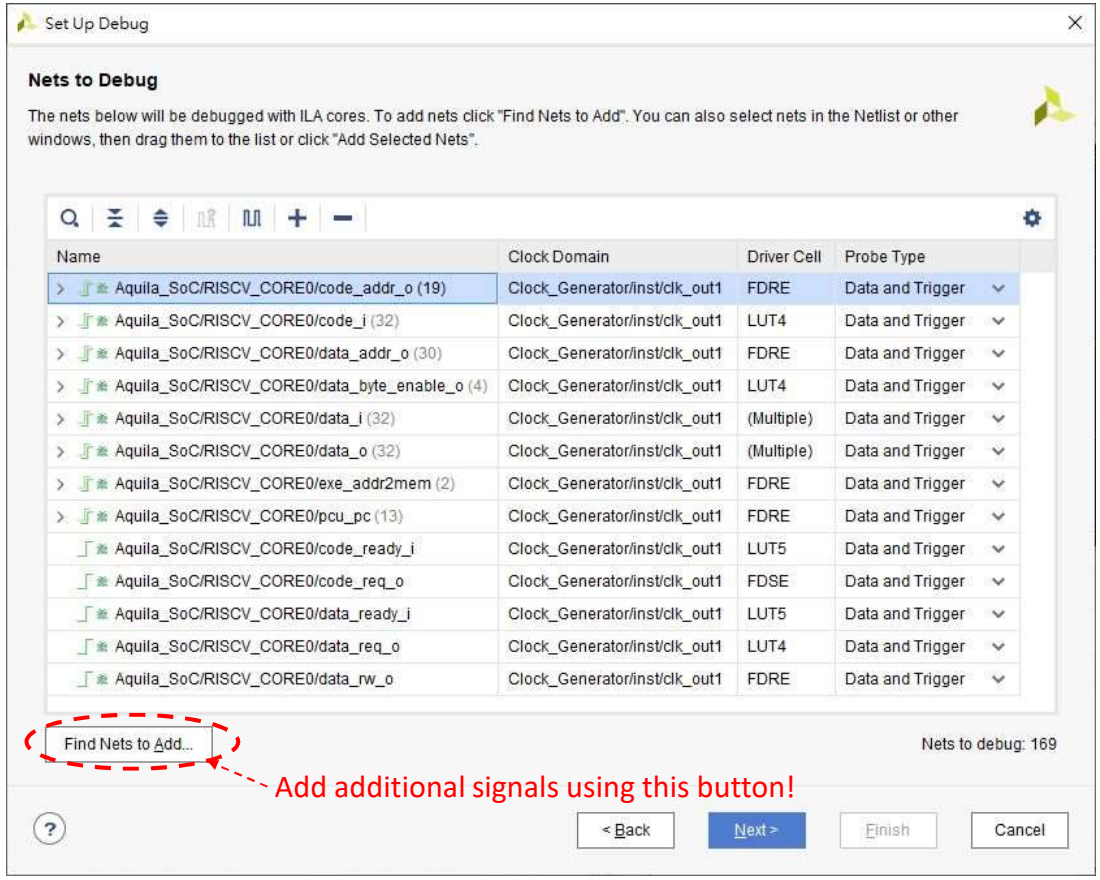
# Set Up the Debug Wizard

- Open the “Set Up Debug” wizard:



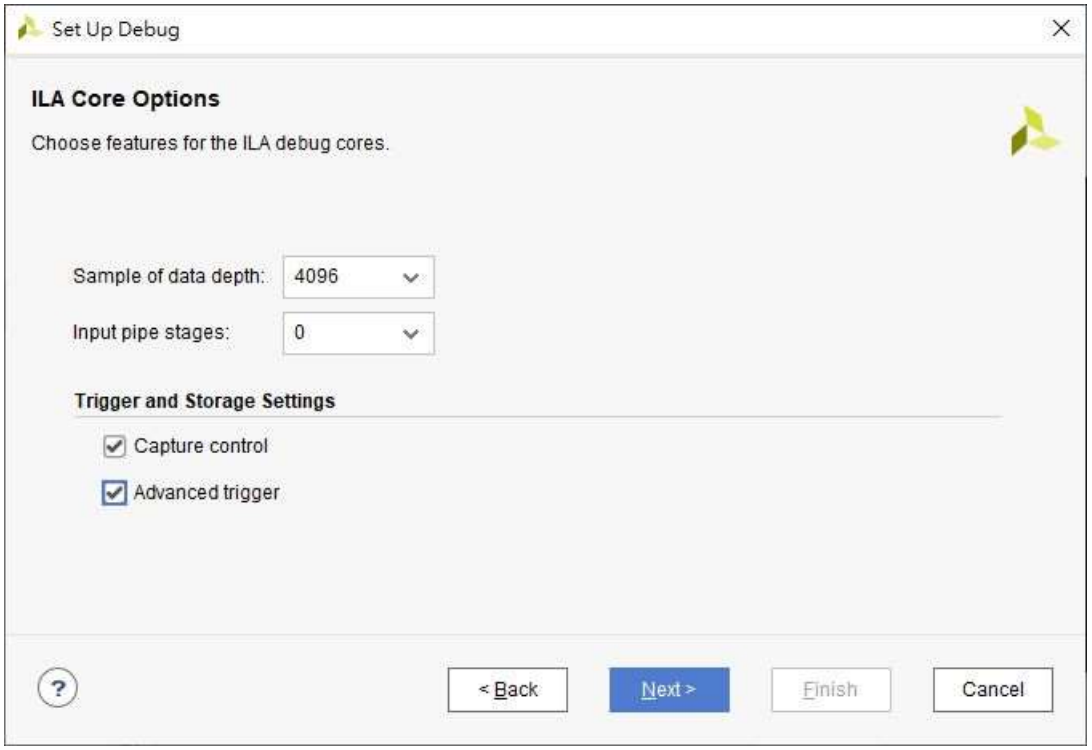
# Double-Check Nets to Be Debugged

- You can add any missing signals in this dialog box
  - Some signals in your Verilog code may be optimized out!

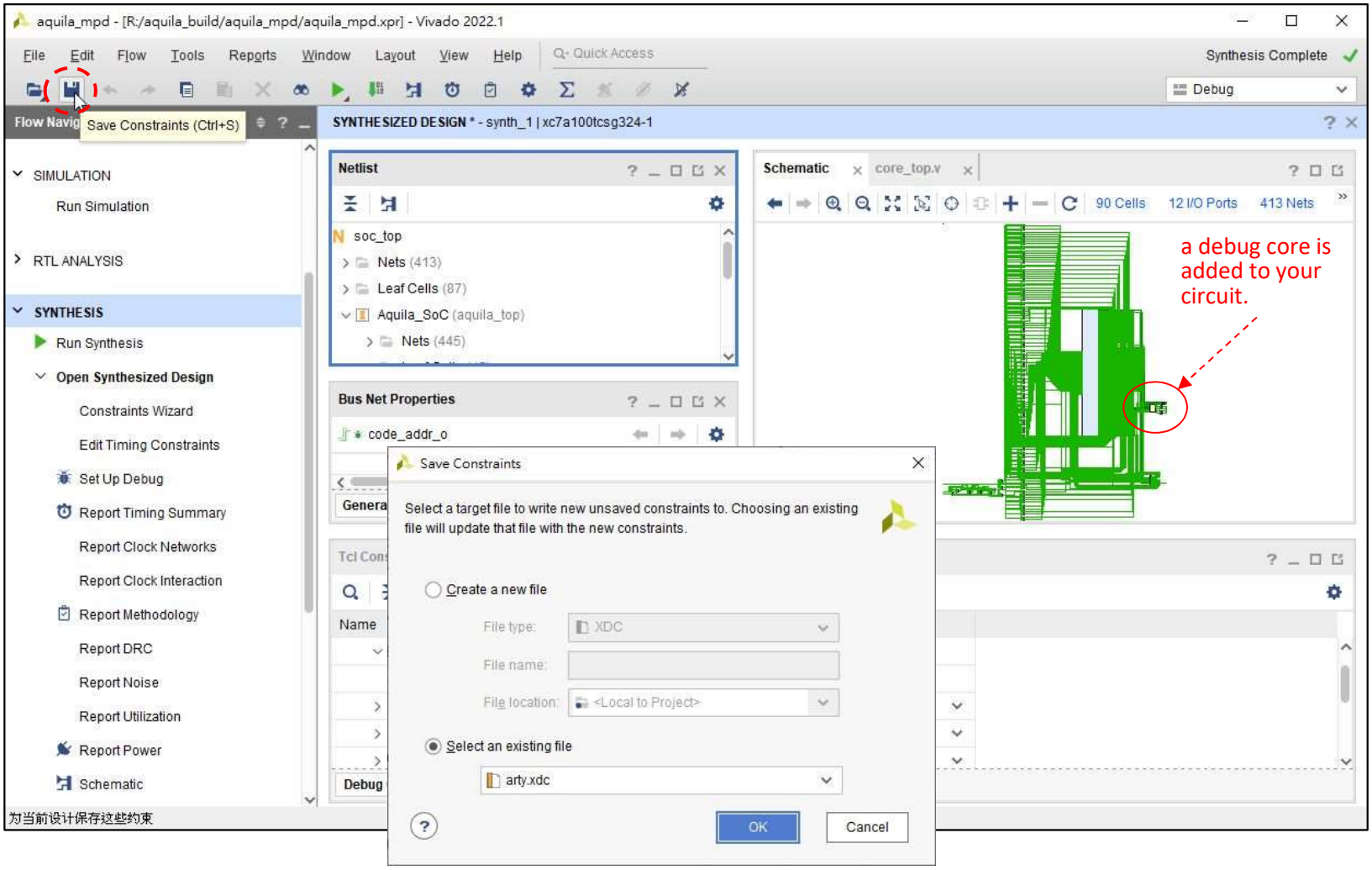


# Modify Trigger Options

- You can check both the “Capture control” and the “Advanced trigger” boxes

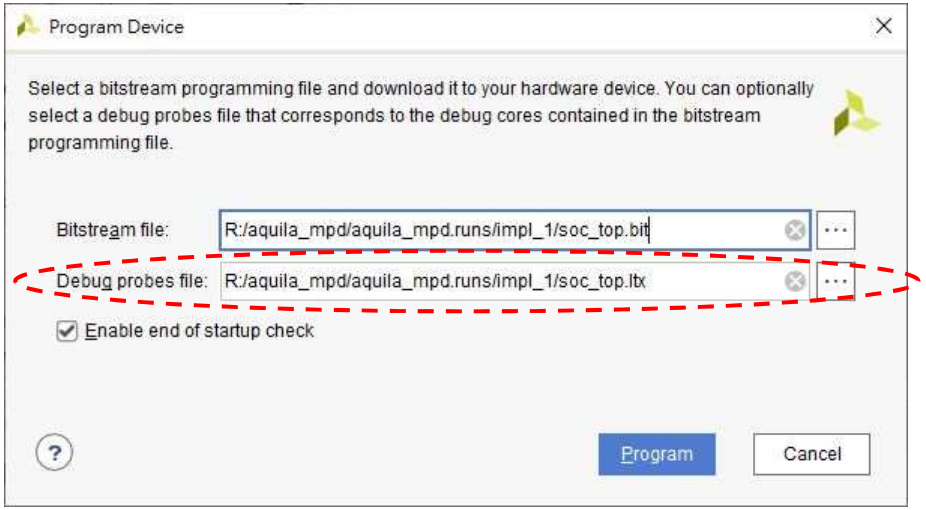


# Save the New Debug Constraints

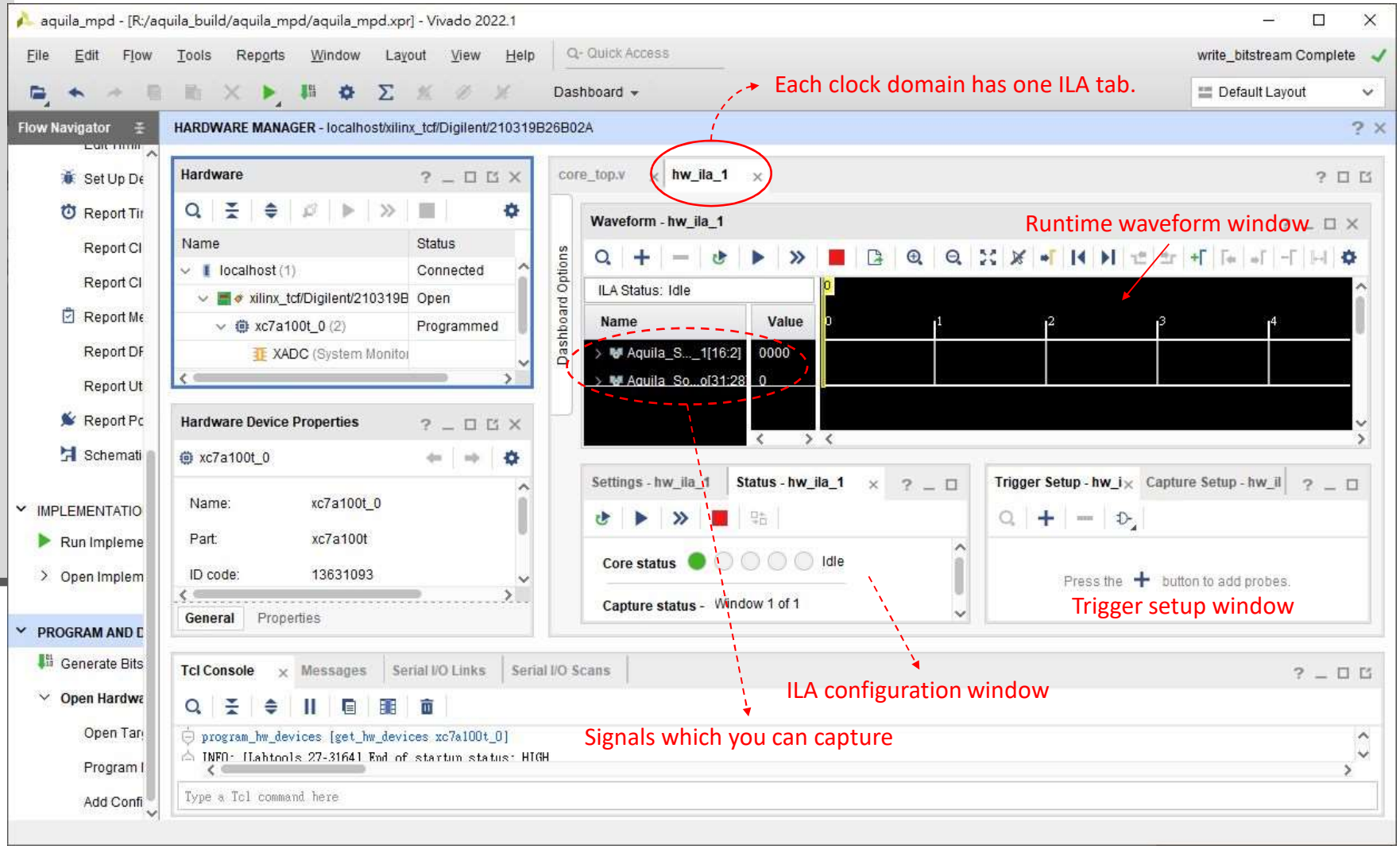


# Program the FPGA for ILA

- After generating the bit file, we can program the FPGA
- Once you hit the “program device” menu item, you will see that an extra ILA configuration file is selected:

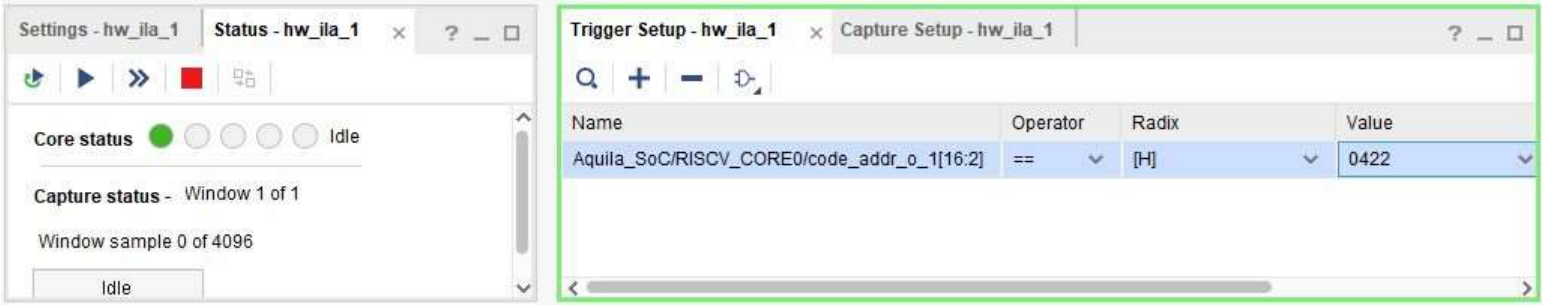


# The Hardware Manager with ILA View



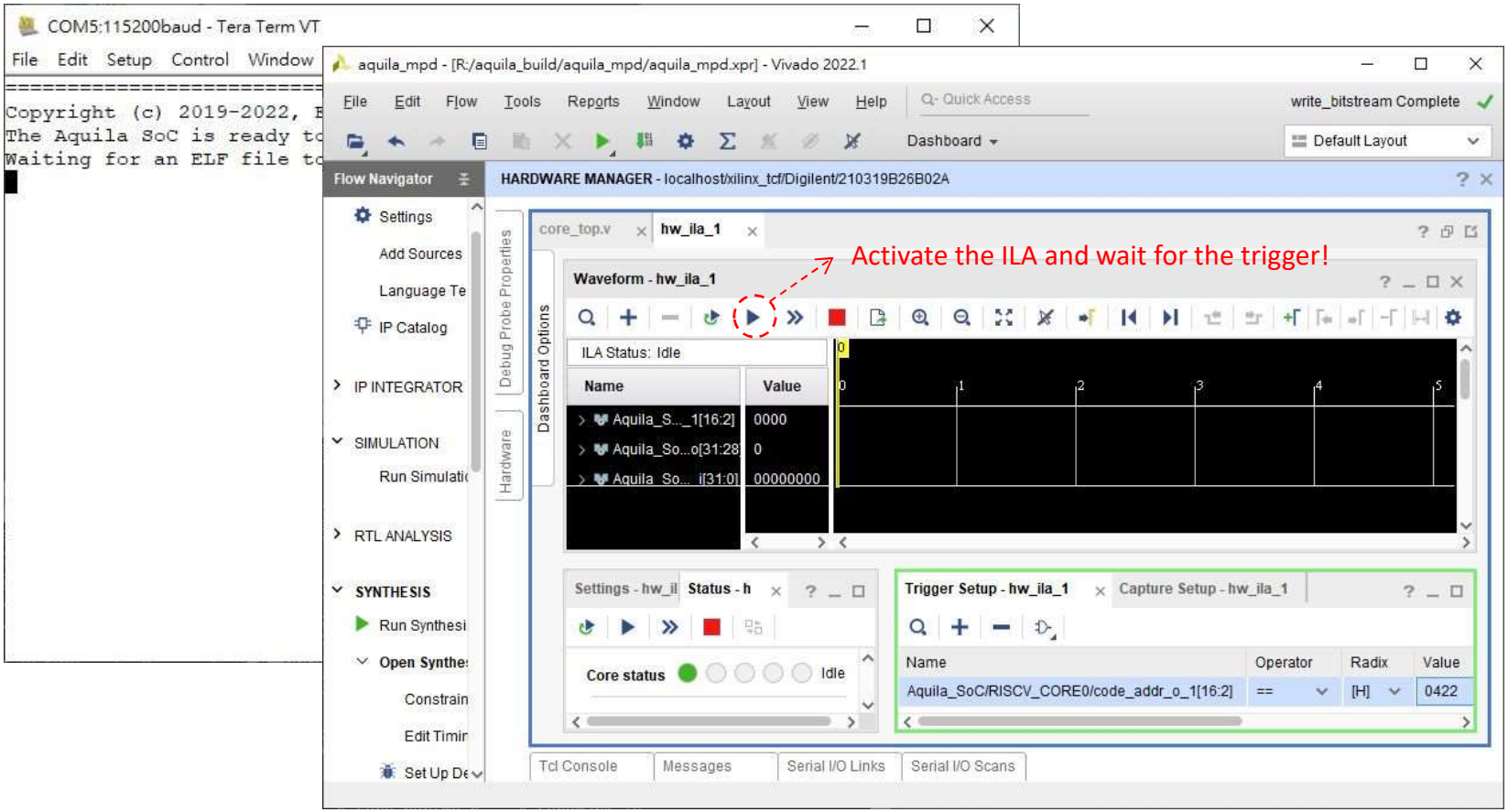
# Setting a Trigger

- A trigger is a signal condition that tells the ILA to begin capturing waveforms
- Set the trigger condition
  - The instruction address is `code_addr_o[31:0]` in `core_top.v`
  - If the `main()` is at `0x1088`, we trigger the ILA when `code_addr_o[16:2]` equals `0x422`.

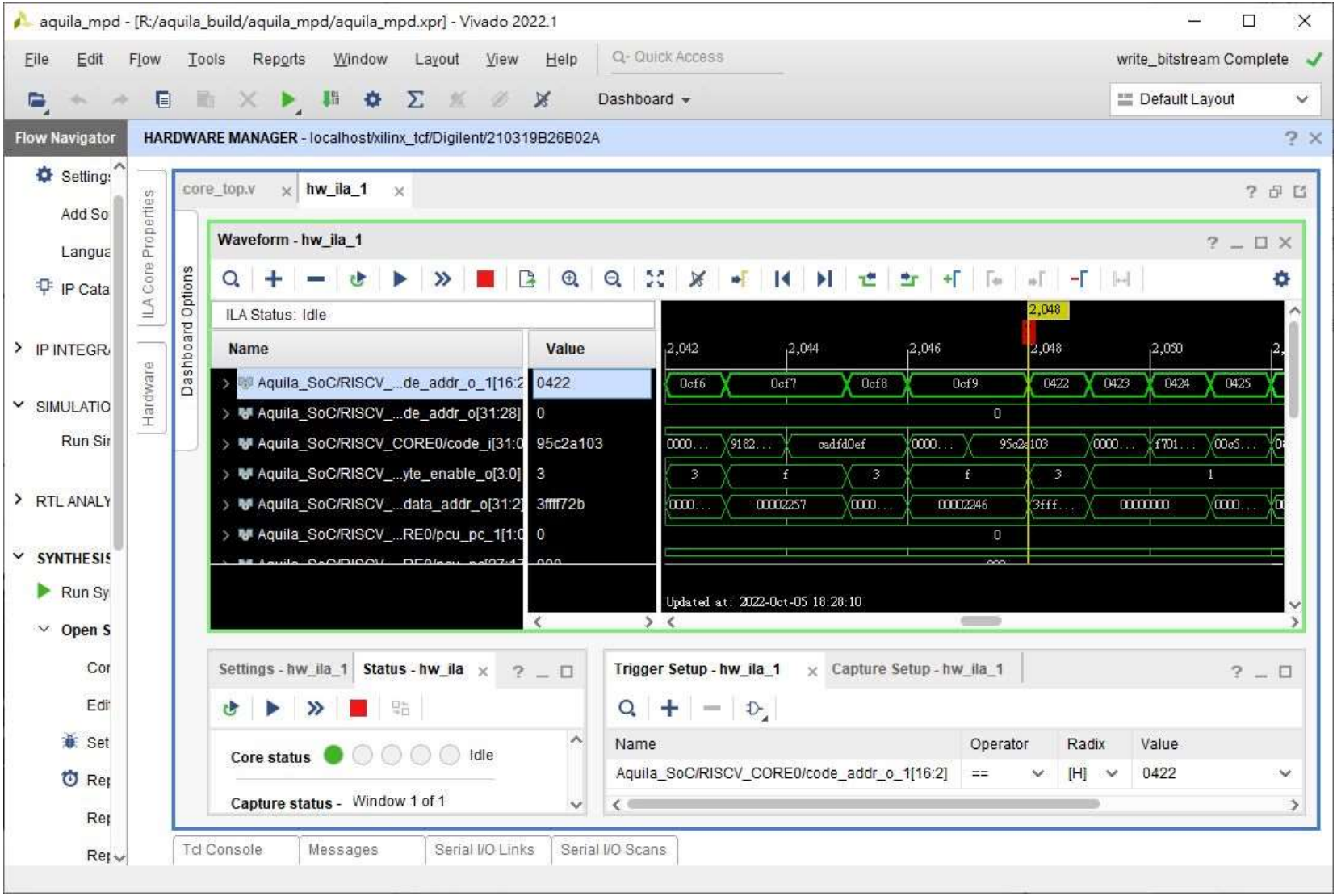


# Capturing the Signals

- Now, you can activate the ILA, then send a program from the UART to FPGA to capture the waveform



# Analyze the Captured Waveform



# Store a Trigger Value in Registers

- In ILA, you can see waveforms of registers easily, but not so easy to check C variables
- You can use inline assembly code to store a C variable into a CPU register so the ILA can display its value:
  - Can be used to set a trigger point!
  - Note, the local variable is assumed to be stored in a register

```
int var1 = 123;

void main()
{
    int var2 = 456;

    /* Save a global variable to register t1 */
    asm volatile ("lui t0, %hi(var1)");
    asm volatile ("lw  t1, %lo(var1)(t0)");

    /* Save a local variable to register t1 */
    asm volatile ("addi t1, %1, 0" : "=r"(var2) : "r"(var2));
}
```